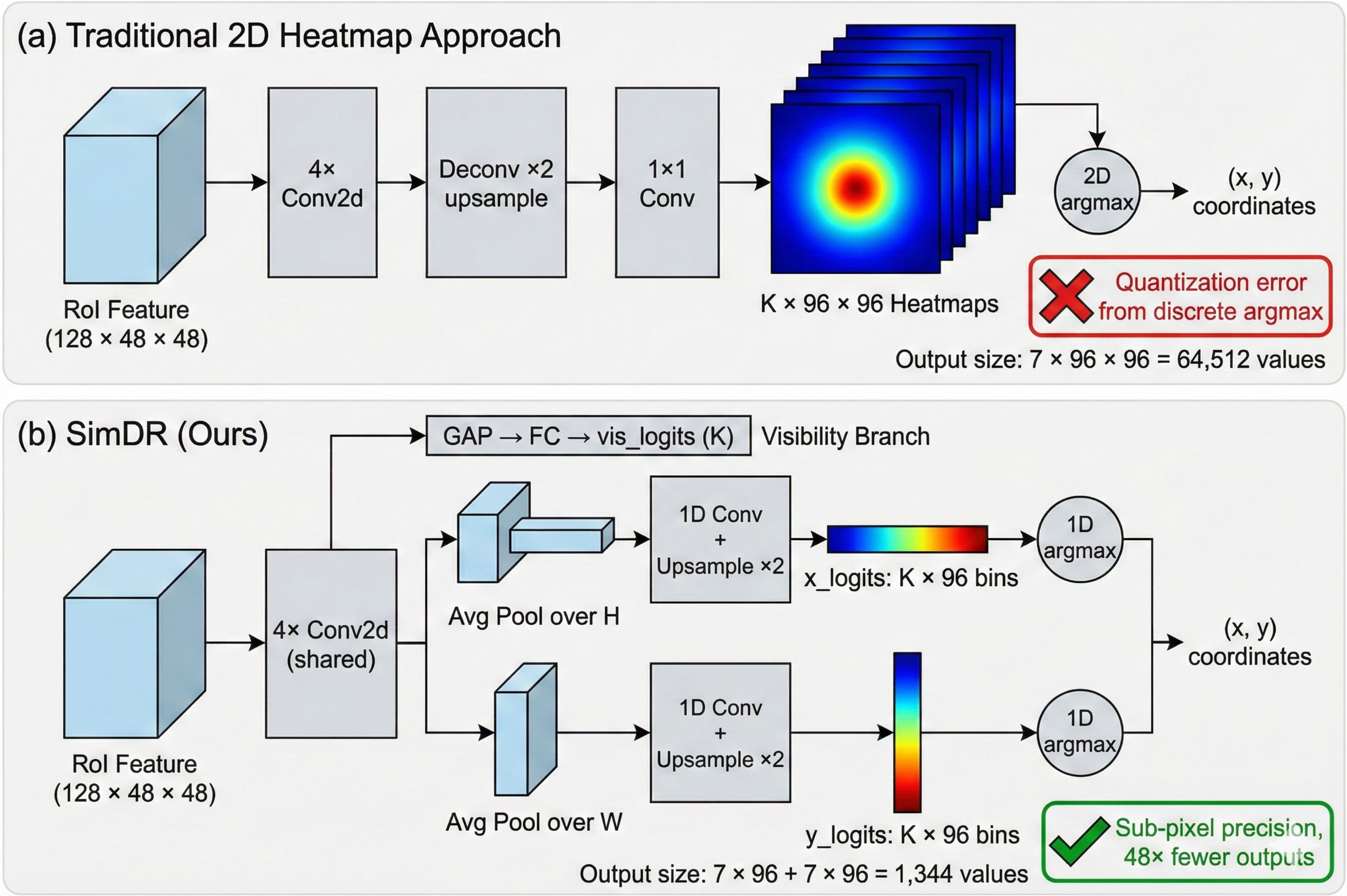
(a) Traditional 2D Heatmap Approach
RoI Feature
(128 × 48 × 48)
4× Conv2d
Deconv ×2 upsample
1×1 Conv
K × 96 × 96 Heatmaps
2D argmax
(x, y) coordinates
Quantization error from discrete argmax
Output size: 7 × 96 × 96 = 64,512 values
(b) SimDR (Ours)
GAP → FC → vis_logits (K)
Visibility Branch
RoI Feature
(128 × 48 × 48)
4× Conv2d (shared)
Avg Pool over H
1D Conv + Upsample ×2
x_logits: K × 96 bins
1D argmax
Avg Pool over W
1D Conv + Upsample ×2
y_logits: K × 96 bins
1D argmax
(x, y) coordinates
Sub-pixel precision, 48× fewer outputs
Output size: 7 × 96 + 7 × 96 = 1,344 values